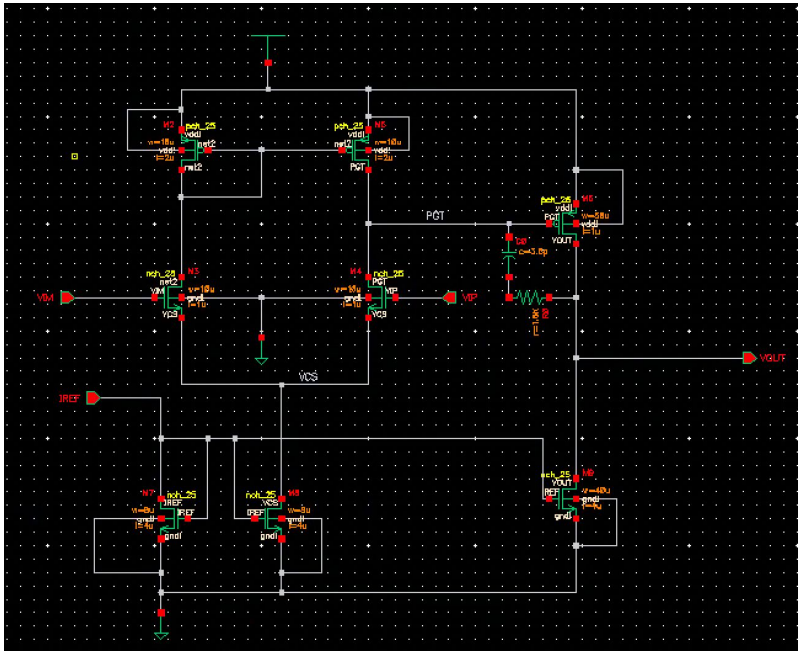
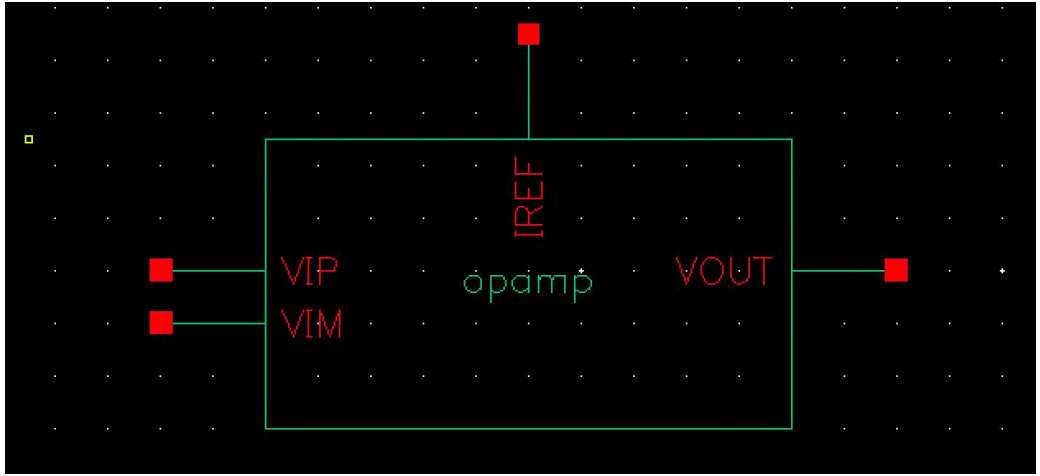
Instrumentation Amplifier: Phase Three

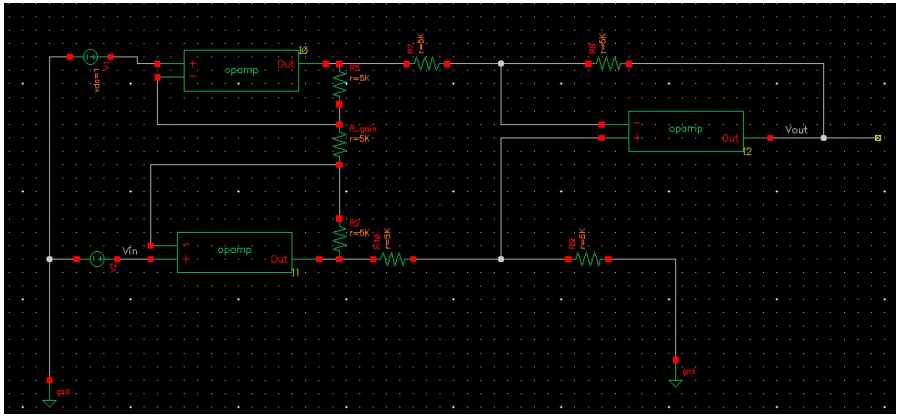
Dylan Thornburg and Idrees Al-Khalidi

Differential Amplifier Schematic and Symbol

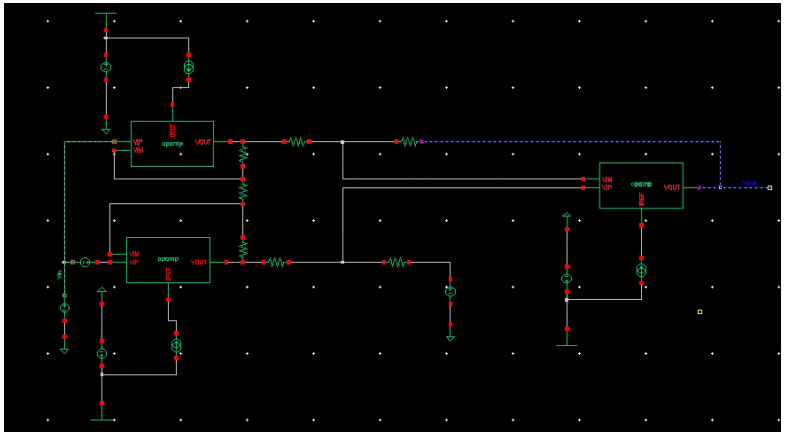




Replacing Ideal Op-Amps with Realistic Op-Amps

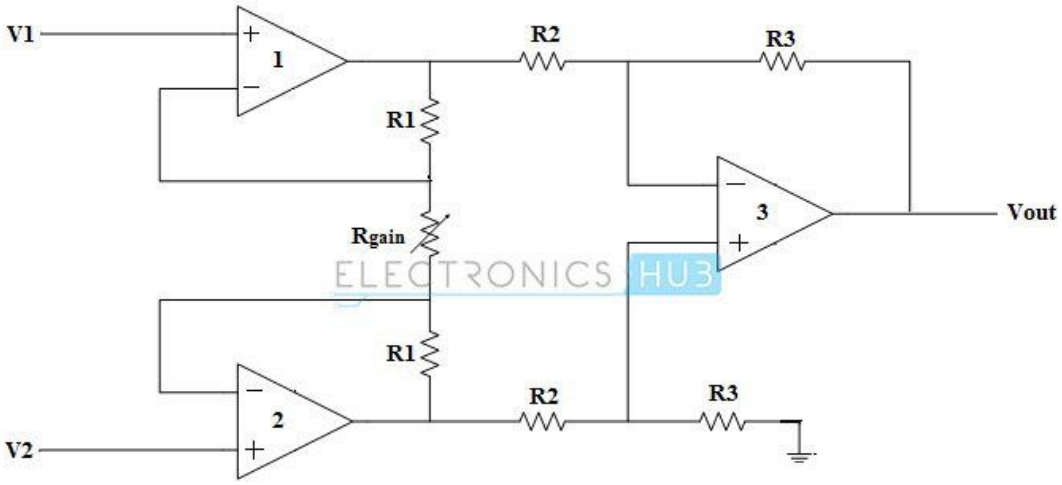


→



Offset Voltage

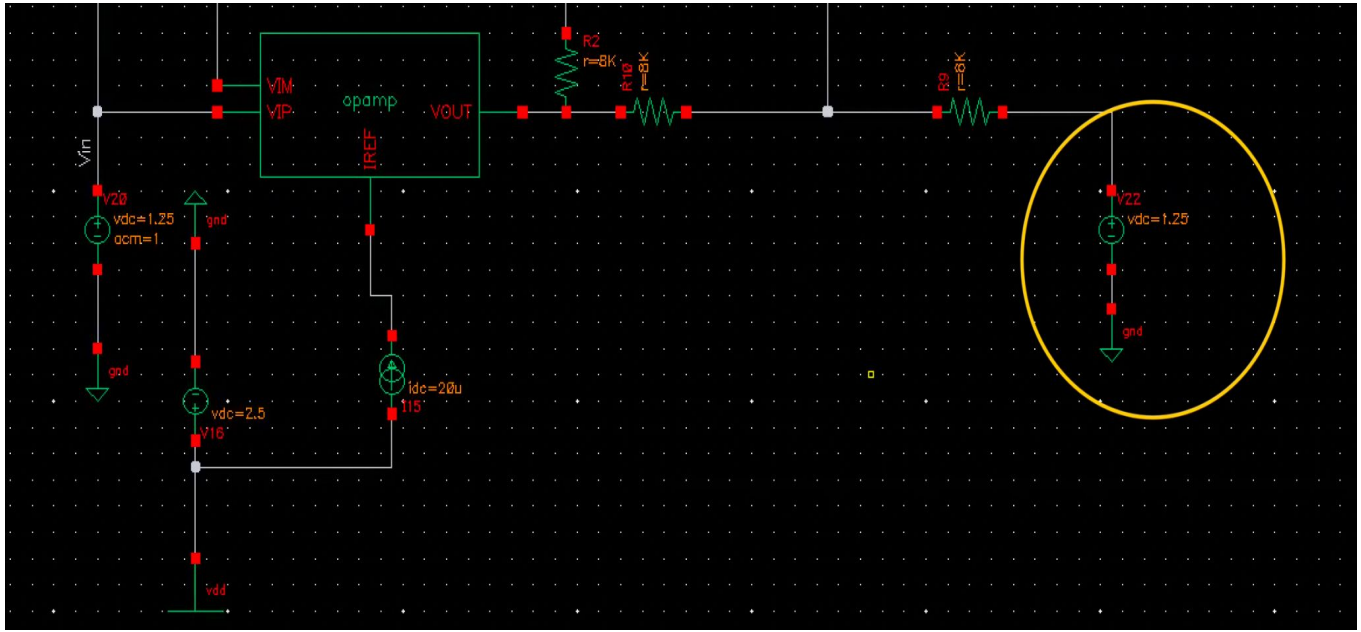
- Voltage of $V_{DD}/2$ (equal to 1.25 V) added to replace ground near R3 by the positive end of Op-Amp 3 in the diagram below to counteract the offset of the common-mode output (schematic shown on next slide)



A Typical Three Op-amp Instrumentation Amplifier

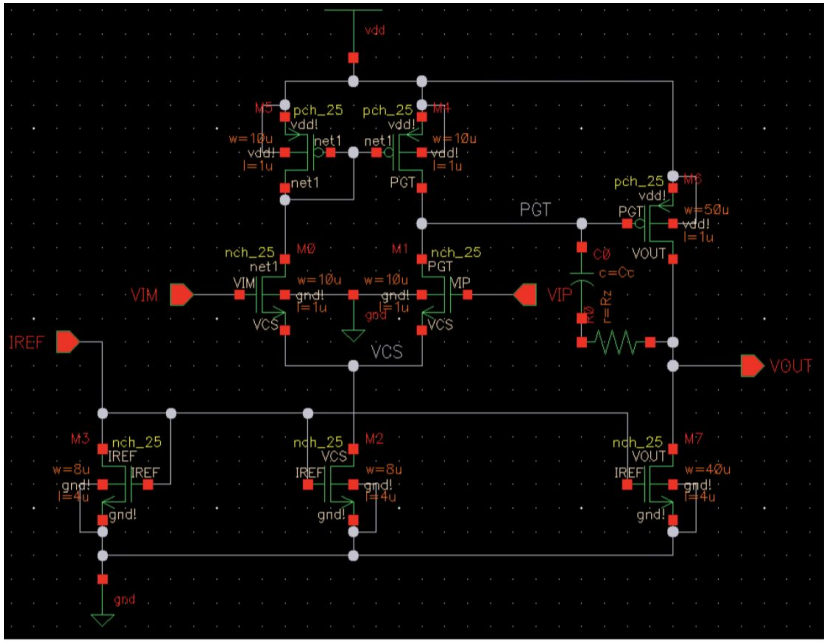
Offset Voltage (continued)

Offset voltage circled on schematic



Input Voltage Range

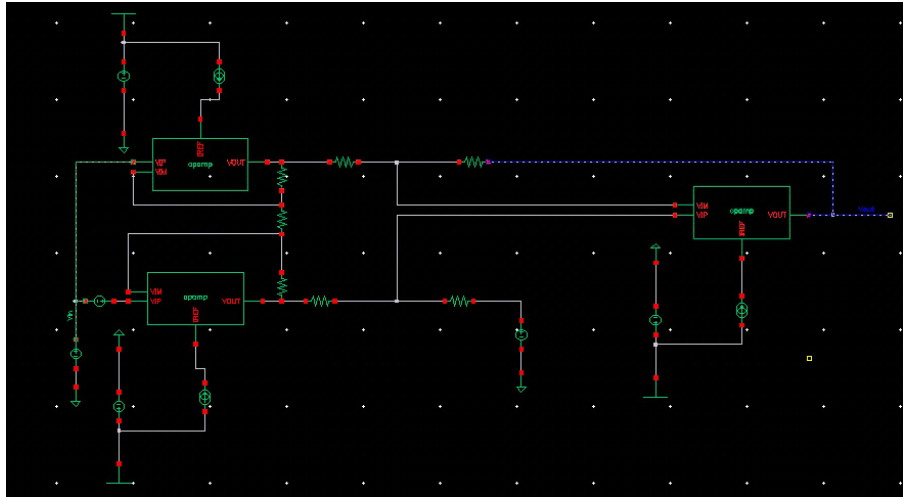
- $v_{I,CM} > V_{ov,2} + V_{GS,2}$
  - (Saturation) $V_{ov,2} = \sqrt{[(2ID_2)/(k_N'(W/L)_7}$ = 0.32 V
  - $V_{GS,2} = V_{ov,2} + V_{T,2}$ = 0.32 V + 0.50 V = 0.82 V
    - $v_{I,CM}$ > 1.14 V (minimum value)
- $v_{I,CM} < V_{DD} - V_{SG,3} + V_{T,1}$
  - V_{DD} = 2.5 V
  - (Saturation) $V_{ov,3} = \sqrt{[(2ID_3)/(k_P'(W/L)_3]}$ = 0.20 V
  - $V_{SG,3} = V_{ov,3} + V_{T,3}$ = 0.20 V + 0.50 V = 0.70 V
  - $v_{I,CM}$ < 2.3 V (maximum value)



Schematic from Lab 7

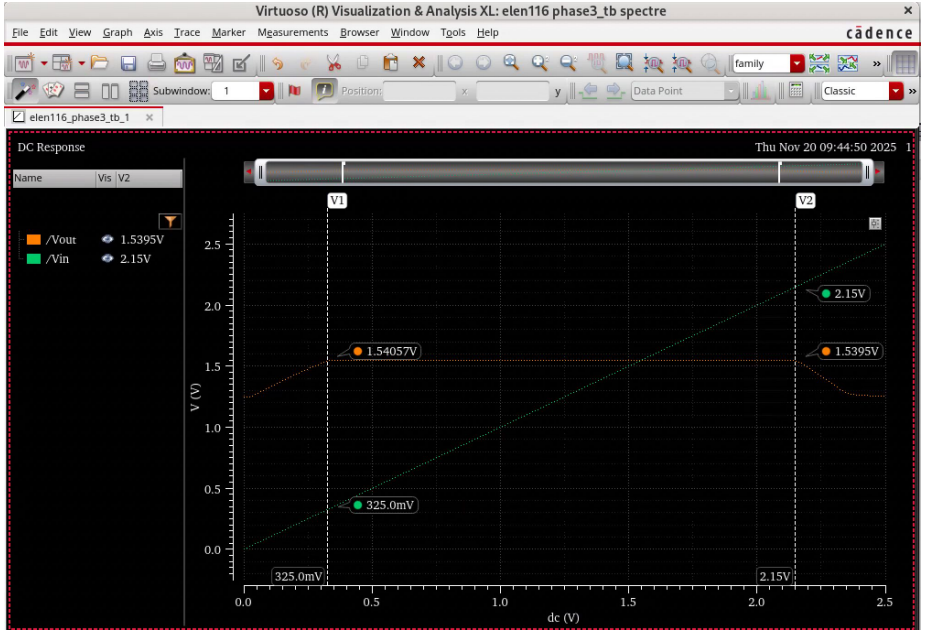
Differential Mode Gain

- If all of the resistors besides R_{gain} are the same value R: $V_{out} = (V_1 - V_2)(1+2R/R_{gain})$
- Ideal: $V_1 - V_2 = 100$ mV, $R_{gain} = R \rightarrow V_{out} = (100\text{ mV})(3) = 300$ mV
  - With offset of 1.25 V $\rightarrow V_{out} = (1.55\text{ V})$
- Common-mode signal is swept from 0 V to 2.5 V and V_{out} is graphed



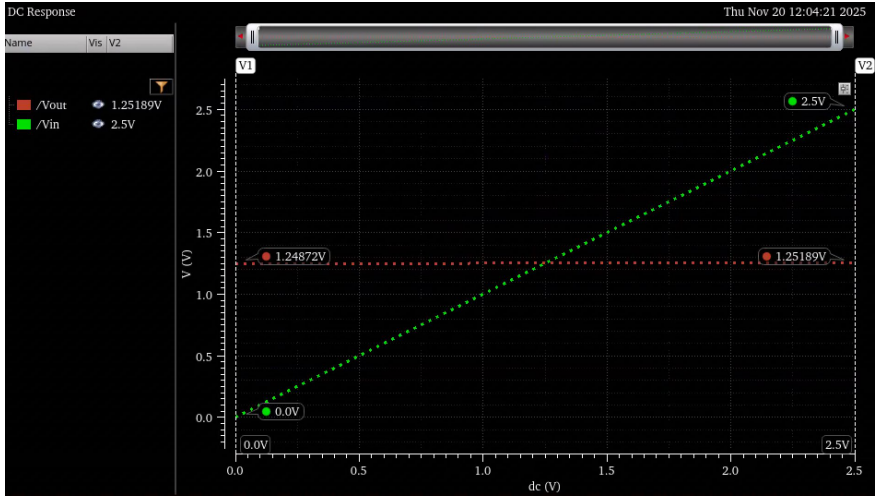
Differential Mode Gain (continued)

- V_{out} = 1.55 V from ~325 mV to ~2.15 V
  - Calculated common mode range: 1.14 V to 2.3 V
  - Non-ideality due to variance in transconductance of MOSFETs and current mirrors not remaining in saturation as the input voltage is lowered



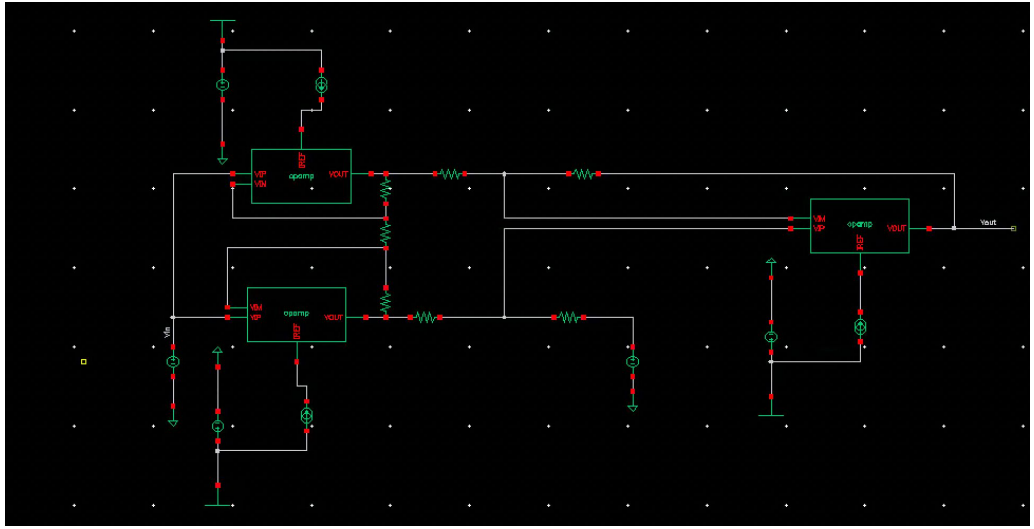
Common-Mode Gain

- Common-mode voltage swept from 0 V to 2.5 V
  - Expected V_{out} = 0 + (offset) = 1.25 V
- Simulation:
  - V_{out} = 1.25 V = Expected V_{out}



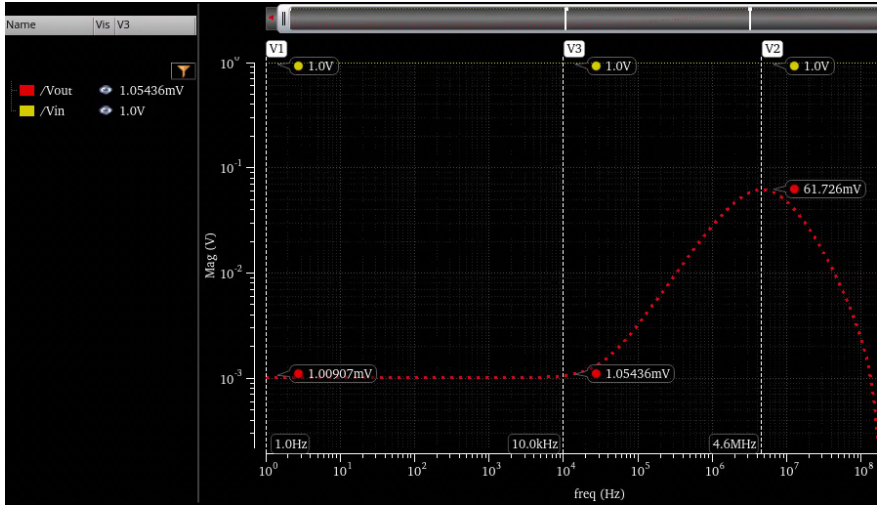
Common-Mode Gain vs. Frequency

- Common-mode input with DC voltage = 1.25 V and AC magnitude of 1 V
    - AC simulation with frequency swept from 1 Hz to 1 GHz



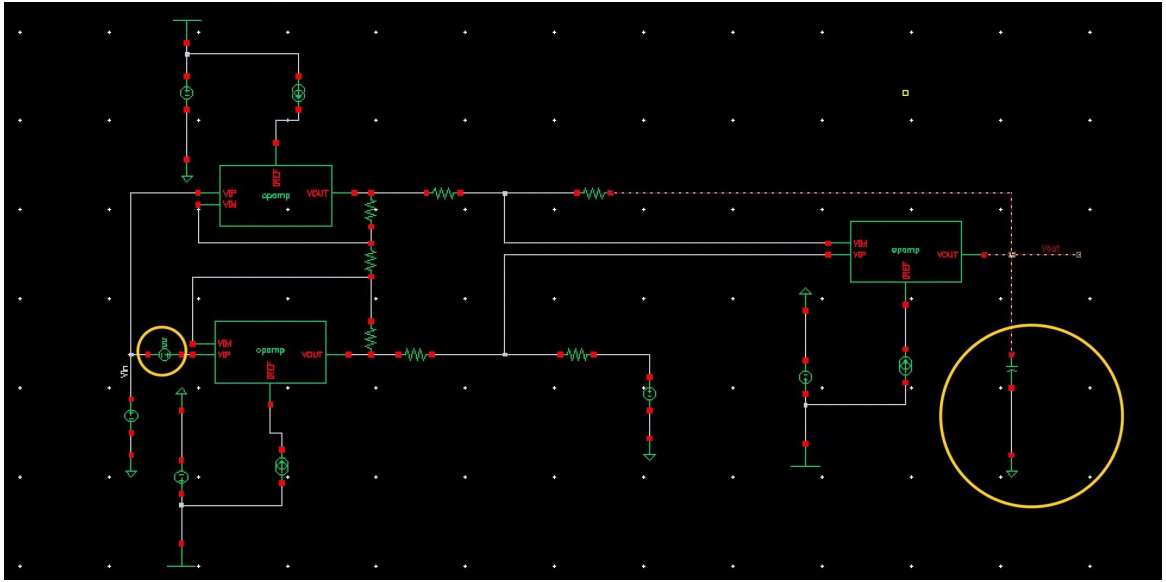
Common-Mode Gain vs. Frequency (continued)

- Common-mode input with DC voltage = 1.25 V and AC magnitude of 1 V
  - AC simulation with frequency swept from 1 Hz to 1 GHz
- Simulation: Input and output are plotted in log scale
  - V_{out} = 0 from 1 Hz to ~10 kHz
  - Real op-amps have finite CMRR that decreases with frequency. In our AC sweep, we reject CM signals up to ~10 kHz, after which internal pole locations in the input stage reduce CMRR. This effect is absent in the ideal Phase 2 model.



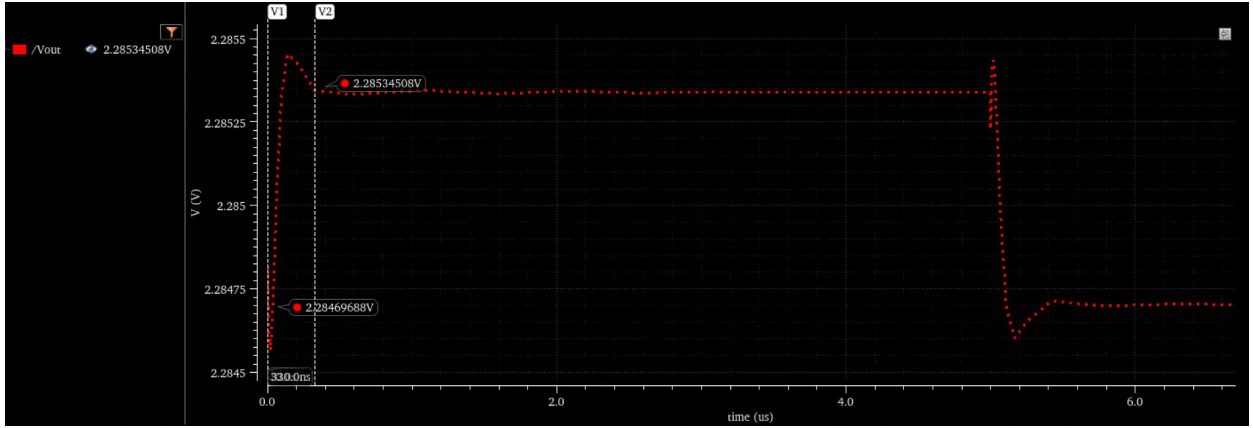
Transient Simulation

- Conditions:
  - Connecting load capacitor of 1 pF to output (circled on schematic)
  - Differential input = pulse wave from 1 V to 2 V with a 10μs period, 1 ps rise time, 1 ps fall time, and a pulse width of 5 μs (50% duty cycle with 10 μs period) (circled on schematic)



Transient Simulation (continued)

- Settling time for V_{out} ~ 330 ns
- The 330ns settling time confirms internal compensation capacitors introduce limited speed, and confirm finite slew rate. This is not an issue for ideal op-amps.



Sources

- https://www.electronicshub.org/instrumentation-amplifier-basics-applications/
- https://www.allaboutcircuits.com/textbook/semiconductors/chpt-8/the-instrumentation-amplifier/#:~:text=What%20Is%20an%20Instrumentation%20Amplifier,adjustment%20of%20multiple%20resistor%20values
- https://www.allaboutcircuits.com/technical-articles/practical-uses-of-instrumentation-amplifiers/
- https://www.youtube.com/watch?v=INeLsIoHw7A